

Distributed Systems

Lecture14: Remote Procedure Call

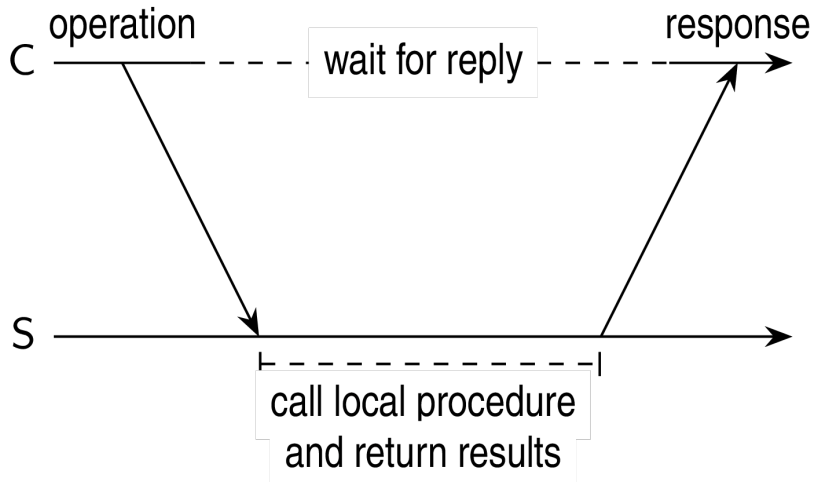
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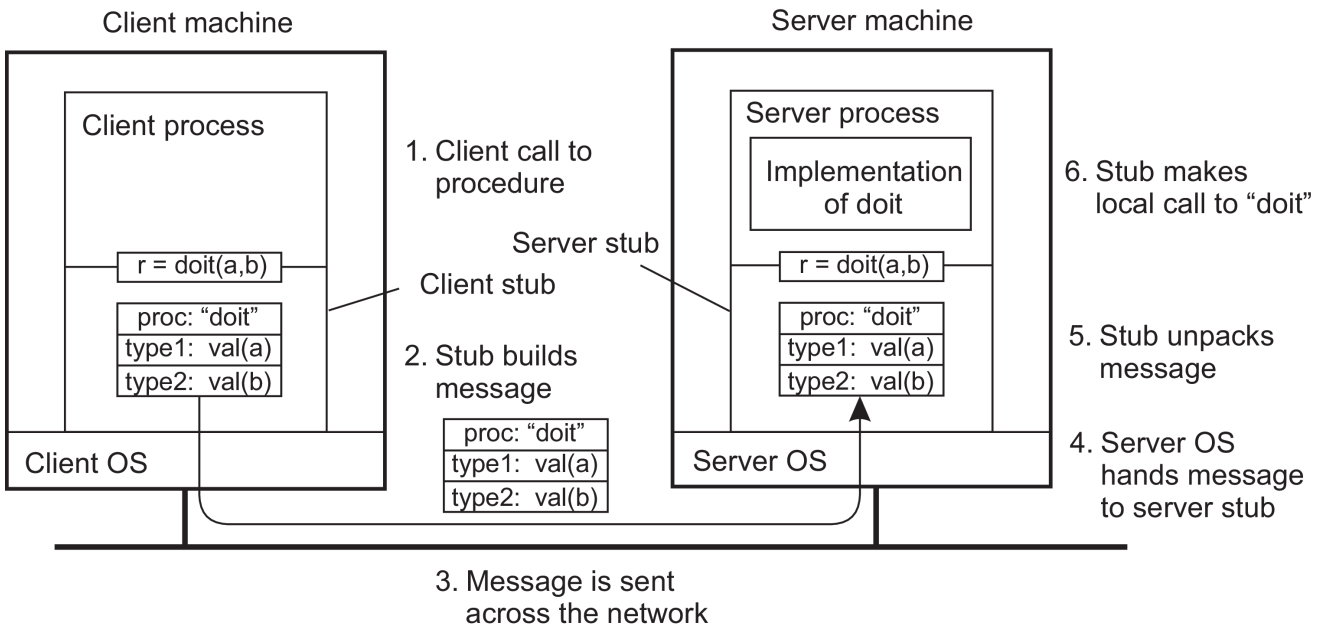
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Basic RPC Operation

- Application developers are familiar with simple procedure model.
- Well-engineered procedures operate in isolation (black box).
- There is no fundamental reason not to execute procedures on separate machine.
- **Communication between caller & callee can be hidden by using procedure-call mechanism.**



Basic RPC Operation



1. Client procedure calls client stub.
2. Stub builds message; calls local OS.
3. OS sends message to remote OS.
4. Remote OS gives message to stub.
5. Stub unpacks parameters; calls server.
6. Server does local call; returns result to stub.
7. Stub builds message; calls OS.
8. OS sends message to client’s OS.
9. Client’s OS gives message to stub.
10. Client stub unpacks result; returns to client.

RPC: Parameter Passing

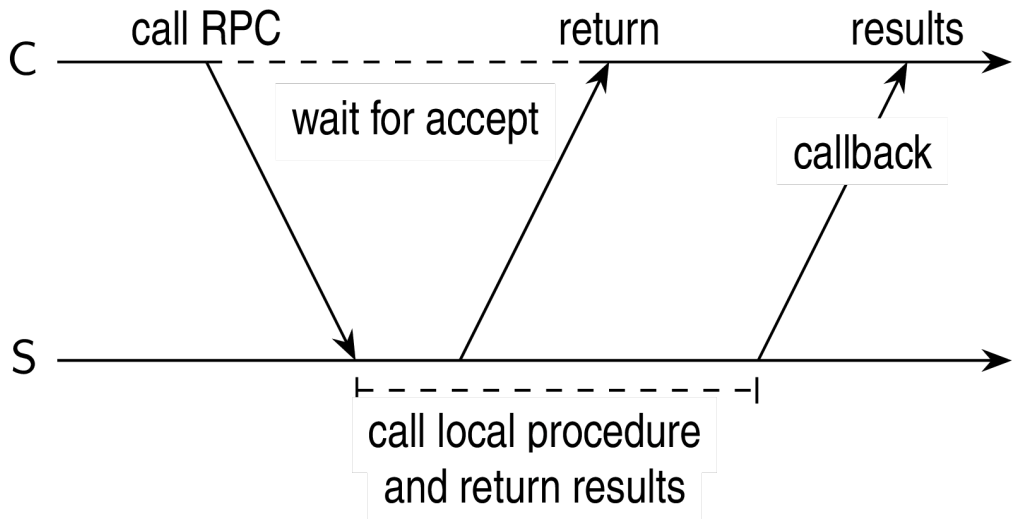
- There's more than just wrapping parameters into a message.
  - Client and server machines may have different data representations (think of byte ordering)
  - Wrapping a parameter means transforming a value into a sequence of bytes
  - Client and server have to agree on the same encoding:
    - How are basic data values represented (integers, floats, characters)
    - How are complex data values represented (arrays, unions)
- **Client and server need to properly interpret messages, transforming them into machine-dependent representations.**

RPC: Parameter Passing

- **Some assumptions:**
  - Copy in/copy out semantics: while procedure is executed, nothing can be assumed about parameter values.
  - All data that is to be operated on is passed by parameters. Excludes passing references to (global) data.
- **Conclusion:**
  - Full access transparency cannot be realized.
- A remote reference mechanism enhances access transparency.
  - Remote reference offers unified access to remote data
  - Remote references can be passed as parameter in RPCs
  - Note: stubs can sometimes be used as such references

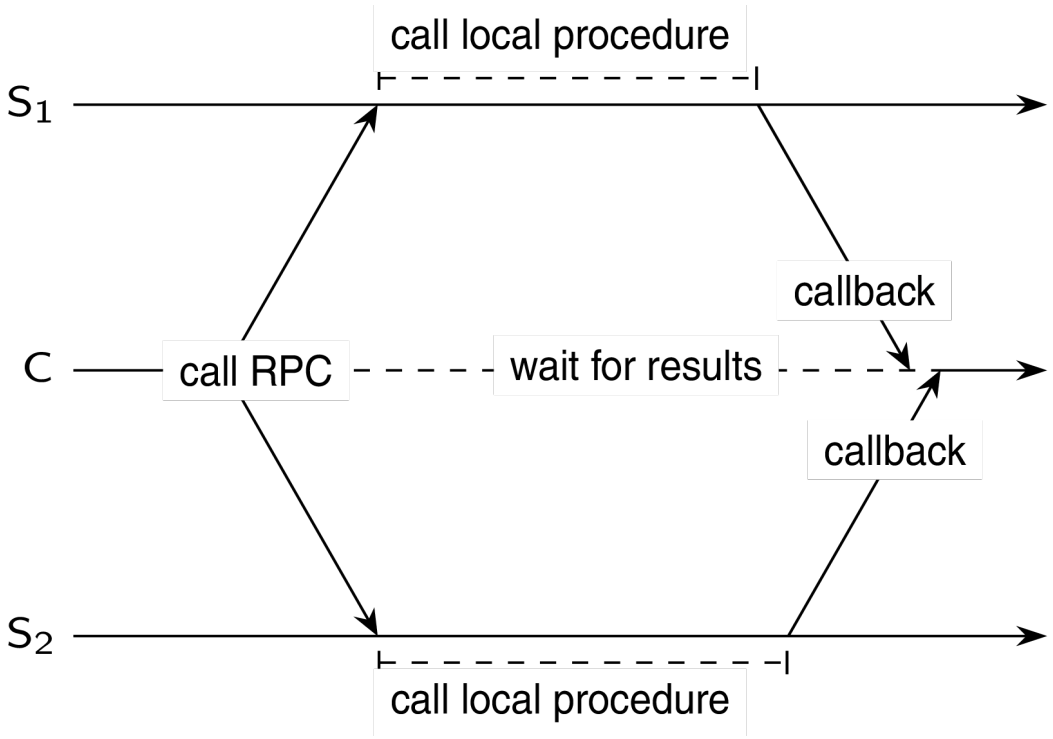
Asynchronous RPCs

- Try to get rid of the strict request-reply behavior, but let the client continue without waiting for an answer from the server.



Sending out Multiple RPCs

- Sending an RPC request to a group of servers.



Wrap-Up + What's Next

In the next lecture, we will discuss the Message-oriented Communication topic, including:

- Transient Messaging with Sockets,
- Persistent Messaging.